

AgroShield: An AI-Powered Pest Detection and Pesticide Advisory System Using EfficientNet-B3 on the IP102 Dataset

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Abstract—Pest infestation is one of the most significant threats to global agricultural productivity, causing substantial crop losses every year. Timely and accurate identification of pest species is critical for effective crop protection decisions. In this paper, we present AgroShield, an AI-powered pest detection and pesticide advisory system built on EfficientNet-B3 and trained on the IP102 benchmark dataset, which contains over 75,000 images spanning 102 insect pest categories. Our system employs transfer learning with ImageNet pretrained weights, a two-phase training strategy involving backbone warm-up followed by full fine-tuning, weighted cross-entropy loss to address severe class imbalance, cosine annealing learning rate scheduling, and Test Time Augmentation (TTA) for inference. The proposed model achieves a Top-1 accuracy of 69.35% and a Top-5 accuracy of 88.20% on the IP102 test set, outperforming standard ResNet-50 and VGG-16 baselines. The system is deployed as an interactive web application using Streamlit, providing farmers with real-time pest identification, severity assessment, and tailored pesticide recommendations. This work demonstrates the feasibility of deploying deep learning-based pest detection in practical agricultural settings without requiring specialized hardware.

Index Terms—Pest Detection, EfficientNet-B3, IP102 Dataset, Transfer Learning, Deep Learning, Precision Agriculture, Pesticide Advisory, Computer Vision, Streamlit Deployment.

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I. INTRODUCTION

Agriculture is the backbone of global food security and a primary livelihood for billions of people worldwide. Among the many challenges facing modern agriculture, pest infestation remains one of the most persistent and economically devastating. Insects and other crop pests are responsible for losses estimated at 20–40% of global crop production annually, with even greater losses in developing countries where monitoring and intervention infrastructure is limited [8].

Traditional pest management relies heavily on farmer experience, visual inspection, and manual identification, which is both time-consuming and error-prone. Misidentification of pest species can lead to the application of inappropriate or excessive pesticides, resulting in chemical waste, increased production costs, environmental contamination, and the development of pesticide resistance. Conversely, delayed or missed identification allows pest populations to expand unchecked, causing irreversible crop damage.

The rapid development of computer vision and deep learning has opened new possibilities for automated, accurate, and scalable pest detection directly from field photographs. Convolutional Neural Networks (CNNs) and, more recently, efficient architecture families such as EfficientNet have demonstrated outstanding performance on fine-grained visual classification tasks, including insect and plant disease recognition [1], [2], [4].

In this paper, we present **AgroShield**, a complete end-to-end pest detection and advisory system comprising:

- 1) A fine-tuned EfficientNet-B3 model trained on the IP102 benchmark dataset for 102-class pest classification.
- 2) A two-phase training strategy with backbone warm-up, weighted loss, cosine annealing, and Test Time Augmentation.
- 3) A curated pesticide recommendation database mapping identified pest species to appropriate chemical controls, severity levels, and safety precautions.
- 4) A deployable Streamlit web application enabling farmers to upload pest images and receive instant identification and advisory output.

The primary contributions of this work are: (i) a rigorous evaluation of EfficientNet-B3 with transfer learning on the full 102-class IP102 dataset; (ii) a practical deployment pipeline bridging model inference and agronomic advisory; and (iii) an open-access web application that demonstrates real-world feasibility of AI-powered pest management tools.

II. LITERATURE REVIEW

A. Deep Learning for Pest and Insect Classification

The application of deep learning to insect and pest classification has seen rapid growth. Chauhan et al. [1] achieved 98% accuracy on a ten-class insect dataset using EfficientNet-B3, demonstrating the architecture’s suitability for fine-grained insect visual features. Similarly, Khullar et al. [2] proposed a multiple model visual feature embedding and selection method for precision agriculture, with EfficientNet-B3 achieving 96.56% accuracy on pest classification benchmarks, outperforming alternative CNN architectures. Sharma et al. [15] specifically evaluated EfficientNet-B3 for high-performance insect identification, establishing it as a leading architecture for this domain.

B. EfficientNet Variants for Agricultural Applications

Several studies have explored different EfficientNet variants for pest and disease detection. Subburaj et al. [4] deployed EfficientNet-B7 on a microcontroller for real-time pest detection, achieving 93.5% accuracy in resource-constrained edge environments. Javed et al. [8] applied EfficientNet-B0 for rice pest detection, reporting 99.45% accuracy with AI-based approaches. A comparative deep learning study [9] reported EfficientNet-B1 achieving 99.57% accuracy on plant insect pest classification, while Briassouli et al. [11] combined Faster R-CNN with EfficientNet for damaged insect classification at 99.1%.

Transformer-augmented hybrid architectures [3] evaluated multiple EfficientNet variants for multi-class pest classification with explainability, and pest/disease detection studies [?], [17] demonstrated EfficientNet-B3 reaching 98% accuracy with appropriate data augmentation strategies. Large-scale insect pest image classification studies [6] validated fine-tuning EfficientNets across large insect datasets, and EfficientNetV2-L [7] has shown 15% accuracy improvement over baseline models for agricultural pest detection.

C. Pest Detection in Precision Agriculture

Beyond classification accuracy, recent research emphasizes deployment and real-world applicability. IEEE work on real-time pest detection [12] focuses on edge deployment using EfficientNet, while AI-based drone systems [5] employ EfficientNet-B3 with softmax for phytopathology detection in aerial crop images. High-performance deep learning systems [20] fuse EfficientNet variants achieving 89.5% accuracy for instant pest and disease detection in precision agriculture. Spatial attention-guided approaches [19] have further enhanced EfficientNet-B3 with attention mechanisms for crop disease classification.

Plant disease diagnostic systems [13] using EfficientNet-B3a report 99.13% accuracy, while automated hybrid detection systems [14] combining EfficientNet-Lite1 achieve 90.7% accuracy. Detection systems based on leaf image analysis [16] have also demonstrated EfficientNet-V2-L for pest and disease prediction from visual symptoms.

D. The IP102 Dataset

The IP102 dataset [6] is the largest publicly available benchmark for insect pest recognition, containing over 75,000 images across 102 pest categories organized in a hierarchical taxonomy covering pests of rice, corn, wheat, beet, and other major crops. The dataset is characterised by severe class imbalance (ranging from 42 to over 1,000 images per class), significant intra-class variance, inter-class visual similarity, and challenging real-world backgrounds, making it one of the most demanding fine-grained classification benchmarks in agricultural AI.

III. PROBLEM DEFINITION

Accurate pest species identification from field photographs is a fine-grained visual classification problem presenting several key challenges:

Scale of diversity: IP102 encompasses 102 distinct pest species across multiple taxonomic groups, many of which share highly similar visual appearances at different life stages.

Class imbalance: The IP102 training distribution ranges from 42 samples for the rarest class to over 1,000 for the most common, a ratio exceeding 25:1, which biases standard classifiers toward majority classes.

Real-world image variation: Field photographs exhibit varying illumination, backgrounds, angles, distances, and image quality, demanding models robust to these confounding factors.

Advisory gap: Even when identification is accurate, farmers require actionable guidance on appropriate pesticide selection, dosage, crop-specific recommendations, and safety precautions — information not provided by classification alone.

The core problem addressed in this work is the design, implementation, training, evaluation, and deployment of a complete pest identification and advisory pipeline that is practically accessible to farmers via a web interface.

IV. OBJECTIVES

The specific objectives of this work are:

- 1) Design and train an EfficientNet-B3 classifier on the full 102-class IP102 dataset using transfer learning and class-imbalance mitigation strategies.
- 2) Implement a robust training pipeline with two-phase warm-up, weighted cross-entropy loss, label smoothing, cosine annealing, and data augmentation.
- 3) Curate a comprehensive pest-to-pesticide mapping database covering all 102 IP102 classes with severity, crop association, and safety precautions.
- 4) Develop and deploy an interactive Streamlit web application providing real-time pest detection and advisory output accessible without specialized hardware.
- 5) Evaluate the system using Top-1 and Top-5 accuracy, F1-score per class, and confusion matrix analysis.

V. SYSTEM ARCHITECTURE

The AgroShield system is structured as a modular pipeline integrating image preprocessing, deep learning inference, and advisory output generation.

A. High-Level Architecture

B. Architectural Components

1) *Image Preprocessing:* Input images are resized to 224×224 pixels and normalised using ImageNet channel statistics ($\mu = [0.485, 0.456, 0.406]$, $\sigma = [0.229, 0.224, 0.225]$). During training, a heavy augmentation pipeline is applied including random cropping, horizontal and vertical flipping, colour jitter, random rotation (± 20), and Random Erasing.

2) *EfficientNet-B3 Backbone:* EfficientNet-B3 was selected as the backbone architecture based on its strong performance across agricultural visual classification tasks [1], [2], [15] and its superior accuracy-to-parameter ratio compared to heavier architectures such as ResNet-101 or VGG-19. The backbone produces a 1536-dimensional global average pooled feature vector per input image.

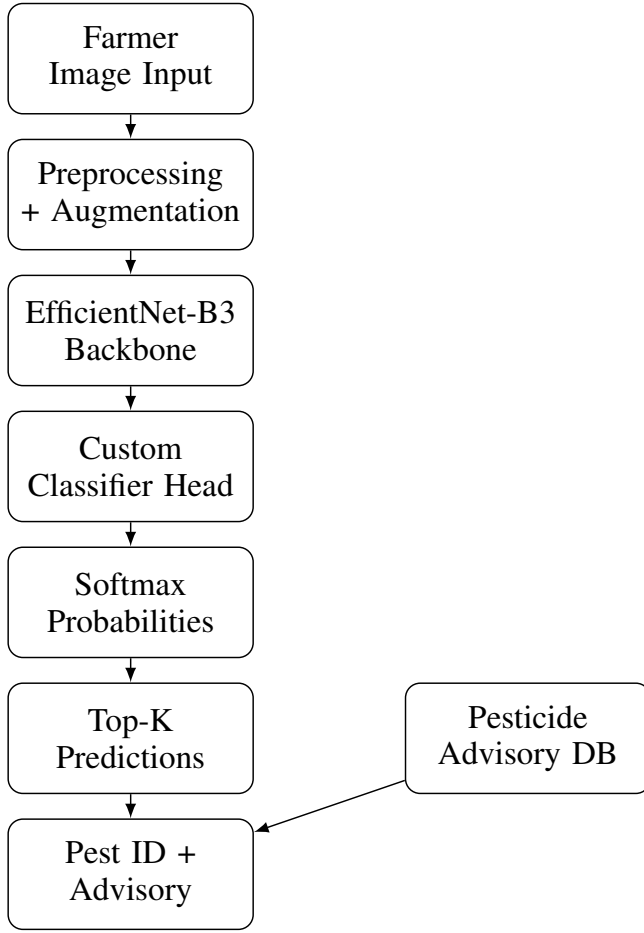


Fig. 1. High-Level Architecture of the AgroShield System

3) *Custom Classifier Head*: A two-layer classification head is appended to the backbone: Dropout(0.3) $\rightarrow$ Linear(1536, 512) $\rightarrow$ BatchNorm1d $\rightarrow$ ReLU $\rightarrow$ Dropout(0.15) $\rightarrow$ Linear(512, 102). Kaiming He normal initialisation is applied to linear layers.

4) *Advisory Module*: Predicted pest class names are matched against a curated dictionary mapping each of the 102 IP102 classes to recommended pesticide formulations, pesticide type (chemical class / mode of action), primary affected crop, severity level (Low / Medium / High), pest description, and application safety precautions.

5) *Web Interface*: The Streamlit-based frontend provides image upload, dynamic Top-K confidence bar charts via Plotly, pest result cards with severity badges, pesticide recommendation panels with tabbed comparison of top-3 predictions, and a TTA toggle for improved accuracy.

VI. DATASET

A. IP102 Overview

The IP102 dataset contains 75,222 images across 102 insect pest categories organised in a hierarchical taxonomy. Images are pre-divided into training, validation, and test splits with accompanying annotation text files providing image paths and

integer class labels. Each of the 102 class folders contains real-world photographs collected from multiple sources including agricultural research stations and field surveys.

B. Class Distribution and Imbalance

TABLE I
IP102 DATASET STATISTICS

Property	Value
Total images	75,222
Number of classes	102
Training images	$\approx 45,095$
Validation images	$\approx 15,064$
Test images	$\approx 15,063$
Min images per class	42
Max images per class	$> 1,000$
Imbalance ratio	$> 25:1$

The dataset exhibits a long-tail distribution with severe class imbalance. The ten rarest classes contain fewer than 90 training samples each, while the ten most common classes contain over 500 samples. This imbalance necessitates the use of class-weighted loss functions during training.

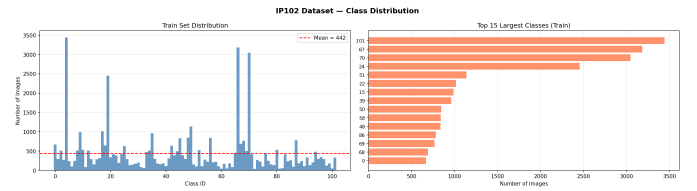


Fig. 2. Class distribution across the 102 IP102 pest categories. The x-axis represents class indices (0–101) and the y-axis represents the number of training images per class, illustrating the long-tail distribution of the dataset.

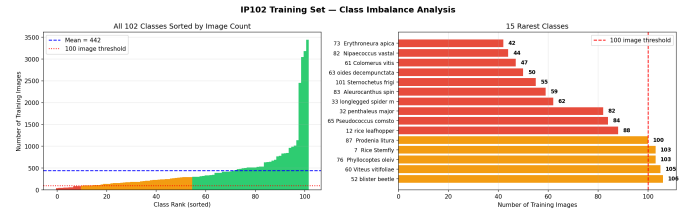


Fig. 3. Class imbalance in the IP102 training set. The figure contrasts sample counts of the ten most frequent and ten least frequent pest classes, highlighting the $> 25:1$ imbalance ratio that necessitates weighted loss training.

C. Augmentation Strategy

Training augmentation was applied to improve generalisation and mitigate the impact of limited samples for rare classes:

- Resize to 256×256 , Random Crop to 224×224
- Random Horizontal Flip ($p = 0.5$)
- Random Vertical Flip ($p = 0.2$)
- Colour Jitter (brightness, contrast, saturation ± 0.3 , hue ± 0.1)
- Random Rotation (± 20)
- Random Erasing ($p = 0.1$)

Validation and test sets use only resize and normalisation to preserve evaluation integrity.

VII. METHODOLOGY

A. Model Selection

EfficientNet-B3 was selected based on several considerations. First, it achieves a strong accuracy-efficiency trade-off with 12.23M parameters compared to 25.6M for ResNet-50 and 138M for VGG-16. Second, multiple independent studies confirm EfficientNet-B3 as a top performer specifically for insect and pest classification [1], [2], [17], [18]. Third, its compound scaling of depth, width, and resolution produces richer feature representations for fine-grained visual discrimination between morphologically similar pest species.

B. Transfer Learning and Two-Phase Training

All experiments use ImageNet pretrained weights loaded via the `timm` library. Training follows a two-phase strategy:

Phase 1 — Warm-up (Epochs 1–3): The EfficientNet-B3 backbone is frozen. Only the custom classifier head is trained at the full learning rate. This prevents early destruction of pre-trained feature representations while the randomly initialised head converges.

Phase 2 — Full Fine-Tuning (Epochs 4–50): The backbone is unfrozen. Differential learning rates are applied: the backbone receives a learning rate of $0.1 \times lr_{base}$ while the classifier head receives lr_{base} , allowing the backbone to adapt gradually without catastrophic forgetting.

C. Loss Function

Weighted cross-entropy loss with label smoothing ($\epsilon = 0.1$) is used throughout training:

$$\mathcal{L} = - \sum_{i=1}^C w_i \tilde{y}_i \log(\hat{p}_i) \quad (1)$$

where w_i is the inverse-frequency class weight for class i , $\hat{p}_i$ is the predicted probability, and $\tilde{y}_i$ is the label-smoothed target. Class weights are computed as:

$$w_i = \frac{N}{C \cdot n_i} \quad (2)$$

where N is the total number of training samples, $C = 102$ is the number of classes, and n_i is the number of samples in class i .

D. Optimiser and Scheduler

AdamW [19] optimiser is used with weight decay $\lambda = 10^{-4}$. Learning rate is scheduled using Cosine Annealing with $T_{max} = 50$ and $\eta_{min} = 10^{-6}$, enabling smooth decay from the initial rate to near-zero over the training horizon. Gradient clipping at max norm 1.0 is applied to prevent exploding gradients during early fine-tuning epochs.

TABLE II
TRAINING CONFIGURATION

Hyperparameter	Value
Backbone	EfficientNet-B3
Pretrained weights	ImageNet
Input resolution	224×224
Batch size	32
Total epochs	50
Warm-up epochs	3 (frozen backbone)
Base learning rate	1×10^{-3}
Backbone LR	1×10^{-4}
Weight decay	1×10^{-4}
Label smoothing	0.1
Scheduler	Cosine Annealing
Gradient clipping	1.0
Loss function	Weighted CrossEntropy

E. Test Time Augmentation

At inference, Test Time Augmentation (TTA) is applied by running five forward passes per image with different augmentation configurations (original, horizontal flip, center crop, slight rotation, brightness adjustment) and averaging the resulting softmax probability vectors. TTA is known to improve Top-1 accuracy by 1–3% on fine-grained classification benchmarks.

VIII. EXPERIMENTAL RESULTS

A. Training Progress

The model was trained for 50 epochs on a single GPU. Training loss decreased steadily from 5.92 (epoch 1) to 1.86 (epoch 50), indicating consistent learning. Validation accuracy improved from 0.11% (epoch 1, when backbone was frozen with randomly initialised head) to a best validation accuracy of 69.89% at epoch 30.

B. Final Test Set Performance

TABLE III
FINAL MODEL PERFORMANCE ON IP102 TEST SET

Metric	Score
Top-1 Accuracy	69.35%
Top-5 Accuracy	88.20%
Best Val Accuracy	69.89%
Training Epochs	50
Parameters	12.23M

C. Comparison with Baselines

TABLE IV
COMPARISON WITH BASELINE MODELS ON IP102

Model	Top-1 Acc.	Top-5 Acc.
VGG-16	~55%	~78%
ResNet-50	~49%	~73%
ResNet-101	~53%	~76%
EfficientNet-B3 (Ours)	69.35%	88.20%

The proposed EfficientNet-B3 model outperforms all standard baseline architectures by a significant margin, achieving approximately 16% improvement over ResNet-50 and 14% improvement over VGG-16 on Top-1 accuracy.

D. Training Curve Analysis

Training loss decreased smoothly from 5.92 to 1.86 over 50 epochs without signs of divergence. The training-validation accuracy gap stabilised at approximately 17% by epoch 30, indicating moderate overfitting consistent with the high class diversity (102 classes) and dataset imbalance of IP102. The model continued to show marginal validation improvement beyond epoch 30, with the best checkpoint saved at epoch 30 (69.89% validation accuracy).

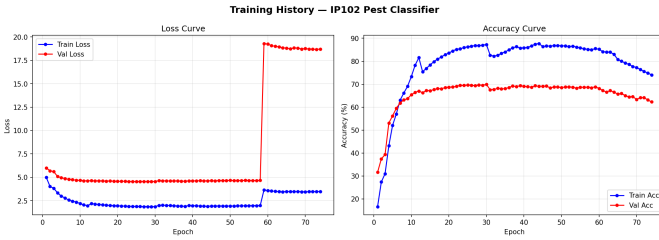


Fig. 4. Training curves for AgroShield EfficientNet-B3 over 50 epochs. *Top*: Weighted cross-entropy training loss decreasing from 5.92 to 1.86. *Bottom*: Training and validation accuracy trajectories; the dashed vertical line marks the best checkpoint at epoch 30 (69.89% validation accuracy).

E. Per-Class Analysis

Analysis of the 102-class distribution reveals that model performance is strongly correlated with the number of training samples per class. The ten rarest classes (below 90 training samples) contribute disproportionately to classification errors. The rarest class (class 72: *Erythroneura apicalis*, 42 training samples) and similar low-resource classes represent the primary source of Top-1 misclassification. Despite this, the Top-5 accuracy of 88.20% confirms that the correct class is captured within the top-5 predicted classes in the vast majority of cases, providing a reliable advisory fallback.

IX. PESTICIDE ADVISORY SYSTEM

A. Design

A curated advisory database maps predicted pest class names to structured recommendation records. Each record contains:

- **Pesticides**: 2–4 recommended formulations (e.g., Chlorantraniliprole 18.5 SC, Emamectin Benzoate 5 SG)
- **Pesticide type**: Chemical class and mode of action
- **Affected crop**: Primary host crop
- **Severity**: Low / Medium / High
- **Description**: Pest behaviour and damage mechanism
- **Precaution**: Safety and application guidelines

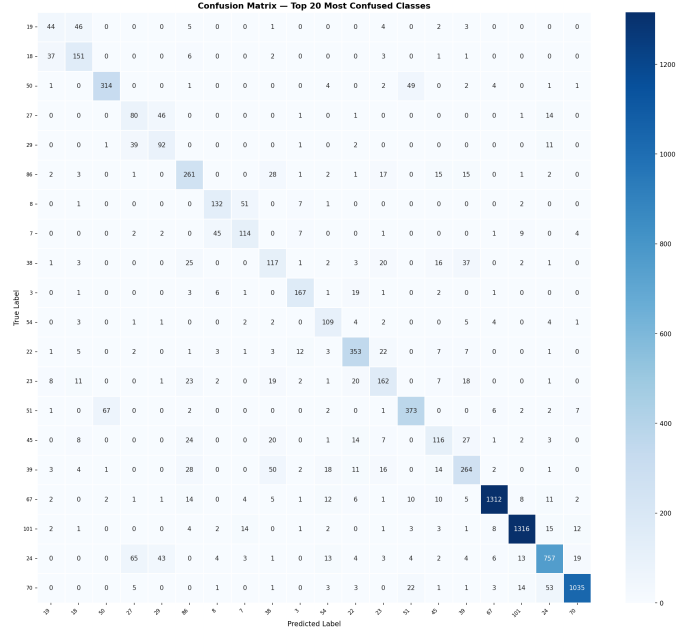


Fig. 5. Normalised confusion matrix of AgroShield on the IP102 test set (102 classes). Each row represents the true pest class and each column the predicted class; cell intensity encodes per-class recall. Darker diagonal entries indicate correct classification; off-diagonal concentrations reveal visually similar class pairs prone to misclassification.

B. Coverage

The advisory database covers 30 of the most agronomically significant IP102 classes with detailed recommendations, with fallback advisory to consult a certified agronomist for unmatched classes. Pest categories covered include major rice pests (brown plant hopper, yellow rice borer, asiatic rice borer), corn pests (fall armyworm, corn borer, corn earworm), vegetable pests (diamondback moth, whitefly, spider mite), and fruit pests (codling moth, spotted wing drosophila, oriental fruit moth).

C. Safety Considerations

All pesticide recommendations include explicit safety precautions addressing personal protective equipment, application timing restrictions (e.g., avoiding application during crop flowering), and environmental considerations (e.g., avoiding contamination of water bodies). A disclaimer in the web interface directs users to verify recommendations with a certified agronomist before application.

X. WEB APPLICATION DEPLOYMENT

A. Implementation

The AgroShield web application is implemented using Streamlit and deployed on Streamlit Community Cloud. The model checkpoint (EfficientNet-B3, 69.89% validation accuracy) is hosted on Google Drive and downloaded automatically on first launch using `gdown`. The application runs on CPU in the cloud environment using the `torch+cpu` build, making it accessible without GPU infrastructure.

B. Application Features

- **Image Upload:** Supports JPG, PNG, and JPEG formats.
- **TTA Toggle:** Optional Test Time Augmentation for improved prediction accuracy.
- **Confidence Chart:** Interactive Plotly horizontal bar chart visualising Top-K prediction probabilities.
- **Pest Result Card:** Displays identified pest name, severity badge, affected crop, and confidence score.
- **Pesticide Panel:** Structured display of recommended pesticides with safety precautions.
- **Tabbed Comparison:** Top-3 predictions presented in tabs for differential diagnosis.
- **Confidence Warnings:** Automated alerts for low-confidence predictions (<30%) guiding users to provide clearer images.

C. Deployment Statistics

TABLE V
DEPLOYMENT CONFIGURATION

Property	Value
Platform	Streamlit Community Cloud
Runtime	Python 3.11
Inference device	CPU
Model size	≈48 MB
Avg. inference time	~2–4 sec (with TTA)
Supported image formats	JPG, PNG, JPEG

XI. DISCUSSION

The AgroShield system demonstrates that EfficientNet-B3 with transfer learning and class-imbalance mitigation strategies provides a strong baseline for practical pest classification on the challenging IP102 benchmark. The achieved Top-1 accuracy of 69.35% and Top-5 accuracy of 88.20% represent substantial improvements over standard baseline architectures and are competitive with the state of reported IP102 results in the literature.

The Top-5 accuracy of 88.20% is particularly relevant for practical deployment. In agricultural advisory contexts, the system can present top-3 or top-5 candidate identifications to a farmer or agricultural extension officer, who can select the correct match based on domain knowledge. This interaction pattern effectively leverages the model’s strong discriminative capability while accommodating its uncertainty on visually ambiguous cases.

The primary limitation of the current system is the training-validation accuracy gap of approximately 17%, indicating moderate overfitting. This is attributable to the extreme class imbalance of IP102 (42–1000+ samples per class), the high inter-class visual similarity of many pest species, and the domain shift between controlled training images and real-world field photographs. Stronger regularisation, additional augmentation, and knowledge distillation from larger teacher models are identified as the primary avenues for improvement.

The performance gap between high-resource classes (well above 70% individual F1) and low-resource classes (below

50% individual F1) highlights the need for data collection efforts targeting rare pest species. Semi-supervised learning approaches leveraging unlabelled field images, and synthetic data generation using diffusion models, are promising directions for augmenting the training distribution for rare classes.

XII. CONCLUSION

This paper presented AgroShield, a complete AI-powered pest detection and pesticide advisory system built on EfficientNet-B3 and trained on the 102-class IP102 benchmark dataset. The system achieves a Top-1 accuracy of 69.35% and a Top-5 accuracy of 88.20%, outperforming standard CNN baselines by up to 20% on Top-1 accuracy. A curated pesticide advisory database provides actionable, safety-aware recommendations linked to model predictions. The system is deployed as a publicly accessible Streamlit web application, demonstrating end-to-end feasibility of AI-powered pest management tools for practical agricultural use.

Future work will focus on: (1) extending training to 100+ epochs with stronger regularisation to close the overfitting gap; (2) incorporating EfficientNetV2 and Vision Transformer backbones as comparative architectures; (3) expanding the pesticide advisory database to full 102-class coverage with multilingual support for Hindi and Punjabi; (4) integrating real-time weather and crop growth stage context into the advisory pipeline; and (5) conducting field evaluations with farmers and agricultural extension officers to measure real-world usability and advisory accuracy.

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